

Yang-Mills flows for Twisted Graphene

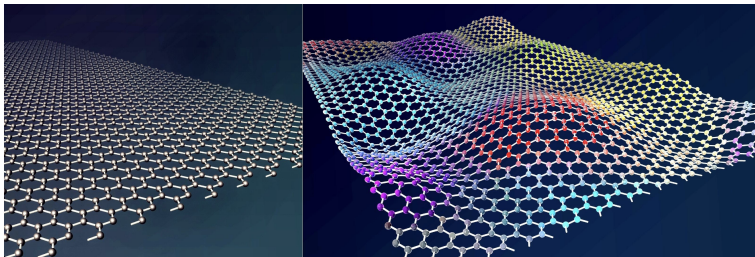
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UC Berkeley, March 18, 2026

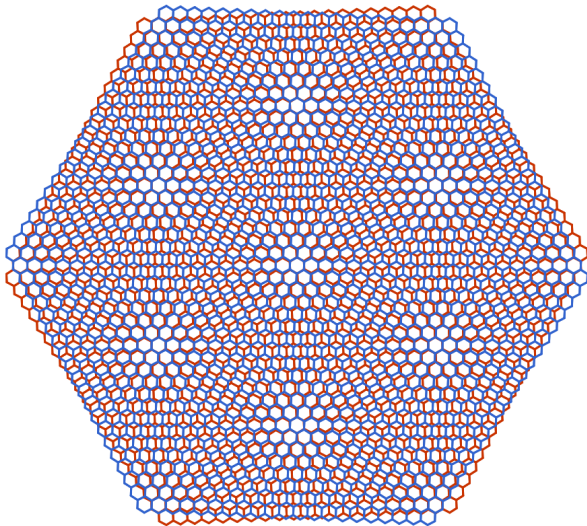
or,

How the work of Narasimhan-Seshadri and Atiyah-Bott
might be useful in condensed matter physics

Joint work with Vasily Iugov (graduate student at YITP)



Moiré period $L(\theta) = \frac{a_0}{2 \sin(\theta/2)}$



Effective Hamiltonian for electron propagation

Lopes dos Santos, Peres, Castro Neto'2007, Bistrizer and MacDonald'2011

in the background of two-layered graphene lattice

can be approximated by a two dimensional Dirac operator

San—Jose, Gonzalez, Guinea'2012

Neglecting interlayer AA coupling gives the chiral limit

Tarnopolsky, Kruchkov, Vishwanath'2019

$$\mathcal{H} = \begin{pmatrix} 0 & \mathcal{D} \\ \mathcal{D}^* & 0 \end{pmatrix}$$

Chiral Dirac operator

$$\mathcal{D} = \bar{\partial}_{\bar{z}} + \mathcal{A}_{\bar{z}}$$

coupled to $SU(2)$ gauge field,

$$\mathcal{A}_{\bar{z}} = \frac{i\alpha}{2} \begin{pmatrix} 0 & U(x, y) \\ U(-x, -y) & 0 \end{pmatrix}$$

determined by a double-periodic (up to a phase) potential $U(x, y)$
with, additionally, a $\mathbb{Z}_6$ symmetry

non-abelian structure due to hopping between layers

It was observed that for “magic” values of the twist angle

$$\alpha = \alpha(\theta)$$

the exactly flat bands are formed: $\mathcal{H}\psi_{k_x, k_y} = 0$

for any quasimomenta k_x, k_y

It was observed numerically that for “magic” values of the twist angle

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Proposals in the literature linking these to superconductivity

Po, Zou, Vishwanath, and Senthil'2018, Yuan and Fu'2018

Our goal: geometric picture explaining this phenomenon

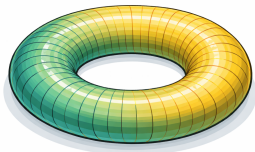
a numerical algorithm for finding ψ_{k_x, k_y}

Our starting point is to observe that the Hamiltonian

$$\mathcal{H} = \begin{pmatrix} 0 & \mathcal{D} \\ \mathcal{D}^* & 0 \end{pmatrix}$$

and its building blocks, chiral Dirac operators $\mathcal{D}$ and $\mathcal{D}^*$

Live on a two-dimensional torus $x, y \sim x + 1, y \sim x, y + 1$



Recall that for **abelian** connection

on a $U(1)$ -bundle over a two-dimensional torus Σ

There is an index theorem:

$$\dim \ker \mathcal{D} - \dim \ker \mathcal{D}^* = \frac{1}{2\pi i} \int_{\Sigma} F_A$$

Also, for generic A either $\ker \mathcal{D}$ or $\ker \mathcal{D}^*$ vanish

For **nonabelian** connection

on a $U(N)$ -bundle over $\Sigma \approx T^2$

There is an index theorem:

$$\dim \ker \mathcal{D} - \dim \ker \mathcal{D}^* = \frac{1}{2\pi i} \int_{\Sigma} \text{Tr } F_A$$

In our story the index identically vanishes, $\text{Tr} F_A = 0$

the existence of zero modes (flat bands) is **non-generic**

How many parameters should one tune for generic $SU(2)$ gauge field?

Hint: existence of zero modes is a feature of a $\mathcal{G}^{\mathbb{C}}$ -orbit

$$\mathcal{A}_{\bar{z}} \mapsto \mathcal{A}_{\bar{z}}^g = g^{-1} \mathcal{A}_{\bar{z}} g + g^{-1} \partial_{\bar{z}} g$$

$$\mathcal{A}_z \mapsto g^{\dagger} \mathcal{A}_z g^{\dagger-1} + (\partial_z g^{\dagger}) g^{\dagger-1}$$

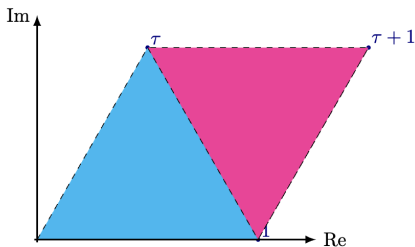
$$z = \alpha + \beta\tau \sim z + m + n\tau, \quad m, n \in \mathbb{Z}$$

In the case of TBG, the associated modular parameter is special

$$\tau = e^{\frac{i\pi}{3}}$$

It corresponds to an elliptic curve with additional symmetry

an orbifold point on $\overline{\mathcal{M}}_{1,1}$



We start with remarks valid for general modular parameter

τ

Role of complex structure: $\mathcal{D} = \partial_{\bar{z}} + \mathcal{A}_{\bar{z}}$

$$\partial_{\bar{z}}(\mathbf{x} + \tau \mathbf{y}) = 0$$

In the context of three-dimensional Chern-Simons theory on $\Sigma \times \mathbb{R}$

The states on Σ (conformal blocks of $\widehat{\mathfrak{g}}_k$)

are functionals of $\mathcal{A}_{\bar{z}}$, obeying

$$\Psi(\mathcal{A}_{\bar{z}}^g) = \Psi(\mathcal{A}_{\bar{z}}) \exp k \left(\int_{\Sigma} \text{tr} g^{-1} \partial_z g \mathcal{A}_{\bar{z}} + S_{WZW}(g) \right)$$

Stability of the $\mathcal{G}^{\mathbb{C}}$ -orbit of $\mathcal{A}_{\bar{z}}$

can be defined **algebra-geometrically**:

there exists a Ψ , obeying, for any $\mathcal{B}_{\bar{z}}$

$$\Psi(\mathcal{B}_{\bar{z}}^g) = \Psi(\mathcal{B}_{\bar{z}}) \exp k \left(\int_{\Sigma} \text{tr} g^{-1} \partial_z g \mathcal{B}_{\bar{z}} + S_{WZW}(g) \right)$$

Such that $\Psi(\mathcal{B}_{\bar{z}} = \mathcal{A}_{\bar{z}}) \neq 0$

Stability of the $\mathcal{G}^{\mathbb{C}}$ -orbit of $\mathcal{A}_{\bar{z}}$

can be defined **differential geometrically**:

there exists a complex gauge transformation g :

$$\text{Such that } F_{\mathcal{A}_{\bar{z}}^g} = 0$$

In other words, the holomorphic bundle, defined by $\mathcal{A}_{\bar{z}}$

can be endowed with hermitian metric,
such that the associated unitary connection is flat

Conformal blocks of $\widehat{su(2)}$
current algebra at positive integer level k

can be studied using free fermions

$$\Psi(\mathcal{A}_{\bar{z}}) \sim \int D\psi D\tilde{\psi} e^{\int_{\Sigma} \tilde{\psi}(\partial_{\bar{z}} + \mathcal{A}_{\bar{z}})\psi}$$

If $\mathcal{A}_{\bar{z}}$ is such, that $\mathcal{D}$ has a zero mode, the Fermi path integral vanishes

Hence we suspect that $\mathcal{G}^{\mathbb{C}}$ -orbit of $\mathcal{A}_{\bar{z}}$ is not stable!

Another, more straightforward, in a sense, argument, is based on

Yang-Mills flow

Another, more straightforward, in a sense, argument, is based on

Yang-Mills flow

$$A = \mathcal{A}_z dz + \mathcal{A}_{\bar{z}} d\bar{z}$$

$$\frac{d}{dt} A = -D_A \star F_A + D_A \chi$$

In other words, we flow
the gauge equivalence class of connection A
in the direction of the gradient of Yang-Mills functional

$$S_{YM} = \int_{\Sigma} \text{tr} F_A \wedge \star F_A$$

Yang-Mills flow

Changes connection A by a complex gauge transformation

$$\frac{d}{dt}A_{\bar{z}} = \partial_{\bar{z}}\Xi + [A_{\bar{z}}, \Xi]$$

The value of S_{YM} decreases

Finite-dimensional analogy

The space of all unitary connections we replace by $\mathbb{C}^N$,
a space of states of N -bit

The group G of unitary gauge transformations we replace by
 $G = U(1)$ - phase redundancy

acting by $(z_1, \dots, z_N) \mapsto (e^{i\theta} z_1, \dots, e^{i\theta} z_N)$

The analogue of the curvature $F_A =$
the moment map for the group action

$$f = \sum_{i=1}^N |z_i|^2 - r$$

The analogue of level k conformal block
A homogeneous degree k polynomial ρ in $z_1, \dots, z_N$

Algebraic geometric stability of $(z_1, \dots, z_N)$:
There exists degree $k > 0$ ρ , such that $\rho(z_1, \dots, z_N) \neq 0$

Differential geometric stability: there exists $g \in \mathbb{C}^\times = G^\mathbb{C}$, such that

$$f(gz_1, \dots, gz_N) = 0$$

Analogue of Yang-Mills flow

$$\frac{d}{dt}\mathbf{z} = -\text{grad}(f^2) = -2f\mathbf{z}$$

Can be integrated analytically

$$\mathbf{z}(t) = e^{-\chi(t)}\mathbf{z}(0), \quad e^{2\chi(t)} = \frac{1 - e^{-4rt}}{r} \sum_{i=1}^N |z_i(0)|^2$$

Thus, for $r > 0$, starting with any $\mathbf{z}(0) \neq 0$, $t \rightarrow +\infty$ limit $\mathbf{z}(t)$ is

$$\mathbf{z}(\infty) = \sqrt{r} \frac{\mathbf{z}(0)}{\|\mathbf{z}(0)\|} \in f^{-1}(0)$$

while $\mathbf{z}(0) = 0$ stays there, $\mathbf{z}(\infty) = 0$, never reaching $f^{-1}(0)$

We have two critical loci of $S = f^2$

$$f^{-1}(0), \text{ where } S_* = 0,$$

and

$$z = 0, \text{ where } S_* = r^2 > 0$$

Back to Yang-Mills theory

Back to Yang-Mills theory

The study of critical points and gradient flows has a rich history

Atiyah–Bott'80, Witten'92, Gross–Matytsin'94,...

Also, fascinating connection to the work'00 of L. Lafforgue

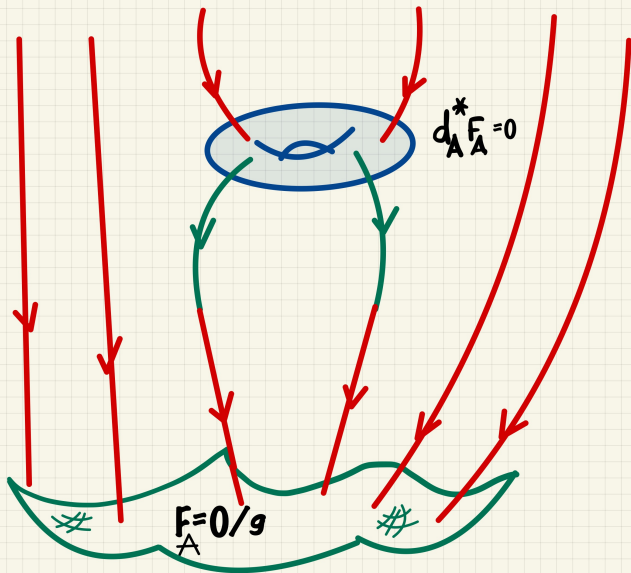
on traces of Frobenius on shtukas cohomology

M. Rappoport, E. Frenkel

Also, related are various proofs of Verlinde formula

Faltings'93, Blau–Thompson'93, Teleman–Woodward'07

A



Critical points of S_{YM} for Σ a two-torus

are easy to classify: either it is an absolute minimum $F_A = 0$ of S_{YM}

or it is a higher critical point

$$d_A \phi = 0, \quad \phi = \star F_A \neq 0$$

where the structure group is reduced to a subgroup commuting with ϕ

Flat connections A with $F_A = 0$ are not good for flat bands
Higher critical points, on the other hand, are good: in the gauge

$$\phi = \text{diag}(\phi_1, \dots, \phi_N)$$

for gauge group $SU(N)$

$$\sum_{i=1}^N \phi_i = 0$$

If the eigenvalues ϕ_i are all distinct

Then the gauge field A is abelian: $A = \text{diag}(a_1, \dots, a_N)$

$$da_i = \phi_i \text{vol}_{\Sigma}, \quad \phi_i = \frac{2\pi i q_i}{\text{Area}_{\Sigma}}, \quad q_i \in \mathbb{Z}, \quad i = 1, \dots, N$$

Higher critical values of Yang-Mills action

$$S_* = \frac{4\pi^2}{Area_\Sigma} \sum_{i=1}^N q_i^2$$

The associated critical locus

Product of $N - 1$ copy of $Jac(\Sigma) \approx \Sigma$

Solutions to chiral Dirac equation

Since some of the $q_i > 0$

The associated Dirac equation has q_i solutions

Turning on quasimomenta k_x, k_y is equivalent to shifts along $Jac(\Sigma)$

Thus we get the desired flat bands

The space $\mathfrak{A}$ of all $\mathcal{A}_{\bar{z}}$ is therefore stratified

$$\mathfrak{A} = \bigcup \mathfrak{A}_{\mathbf{q}}$$

$\mathfrak{A}_{\mathbf{q}}$ is attracted, by the YM flow

to the solution of YM equations

with the vector $\mathbf{q} = (q_1, \dots, q_N)$ of fluxes

For $N = 2$ this is general situation for higher critical points

For $N > 2$ (multilayered multi-twisted graphene) there are mixed phases

There are interesting connections to work of Friedman, Morgan, Witten'97

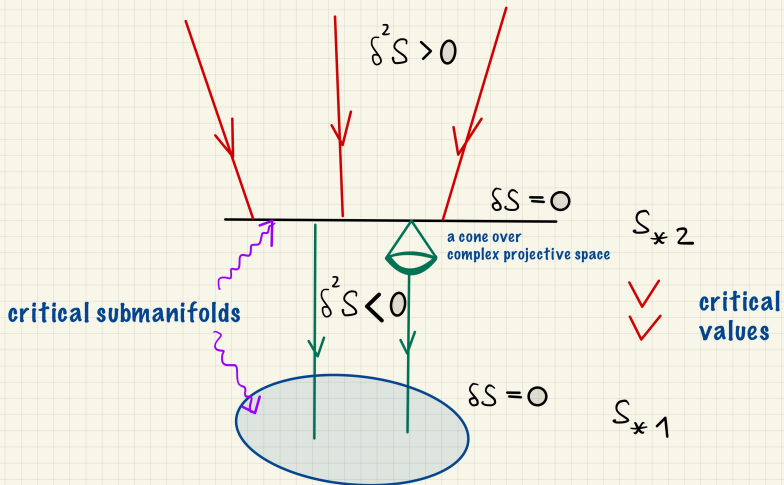
Remaining puzzle: codimension

Generic d -parametric family of connections $\mathcal{A}_{\bar{z}}$

will intersect the attracting manifold of a higher critical point $(q, -q)$

If $d >$ dimension of the repulsive manifold

In the language of renormalization group, one needs to tune to zero the coefficients of relevant and marginal perturbations



The relevant perturbations correspond to the extensions

$$0 \rightarrow L \rightarrow E \rightarrow L^{-1} \rightarrow 0$$

which are classified by cohomology $H^0(\Sigma, L^{\otimes 2}) \approx \mathbb{C}^{2q}$

Equivalently, one can compute the spectrum of $\delta^2 S_{YM}$

Thus, for $q = 1$, one expects to find
 $3 = 4 - 1$ relevant (modulo gauge group) perturbations
and 2 marginal ones

Application

Since the critical values of YM action

$$S_* \geq \frac{8\pi^2}{Area_\Sigma}$$

One can rule out the ansatz for a $\mathcal{A}_{\bar{z}}$

$$\text{If } S_{YM} < \frac{8\pi^2}{Area_\Sigma}$$

We did verify, that the ansatz of *Tarnopolsky, Kruchkov, Vishwanath*
upon the Yang-Mills flow arrives, for (corrected) magic angles
at the $q = 1$ higher critical point

How is it then that just *one* parameter (angle) is tuned
to arrive at the magic angle? In this game $1 \ll 5!!$

The answer is the additional $\mathbb{Z}_6$ symmetry of Σ

The group $\mathbb{Z}_6$ acts on the space of connections on a **trivial $SU(2)$ bundle**

The ansatz of *Tarnopolsky, Kruchkov, Vishwanath* is $\mathbb{Z}_6$ -invariant

The flow commutes with $\mathbb{Z}_6$ -symmetry

The space $V_+ = \mathbb{C}^2$ of relevant deformations decomposes under the action of $\mathbb{Z}_3$

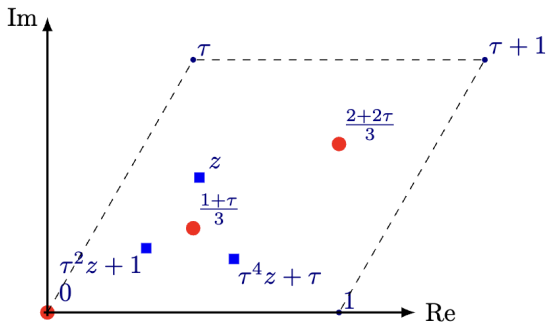
$$V_+ = \mathcal{R}_2 \oplus \mathcal{R}_2^*$$

which means that in the 3-dimensional cone
over $\mathbb{P}(V_+) \approx S^2$,



two 1-dimensional **mustashes** are $\mathbb{Z}_3$ -invariant

Also, in the $Jac(\Sigma)$ of marginal deformations
only three isolated points are $\mathbb{Z}_6$ -fixed



Orbits of $\mathbb{Z}_3$: one-point orbits are represented by red circles and a three-point orbit is represented by blue squares

Thus, the $\mathbb{Z}_6$ -invariant locus has codimension one!

Explaining the magic angle phenomenon

Within the model of Tarnopolsky, Kruchkov, Vishwanath

Future directions

Role of higher critical points? New phenomena for $N > 2$?

On YM side interesting mixed patterns of symmetry breaking are possible

$$SU(N) \longrightarrow SU(N-1) \times U(1)$$

Translating physical quantities of interest to gauge theory context?

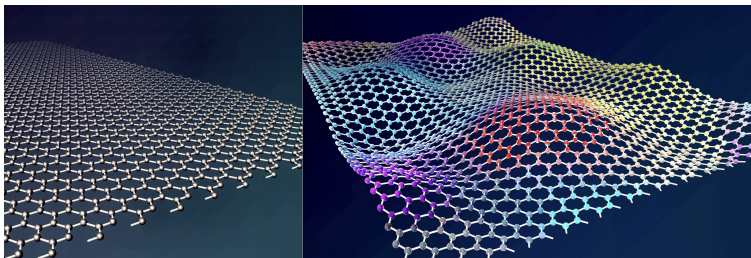
Nearly flat bands for more general model (with AA and AB couplings)?

Yang-Mills flow is a boundary RG flow for open strings –

what is the meaning of this flow in the context of

twisted multilayered graphene?

What would be the analogous gauge theory problem
for more sophisticated graphene sheets?



THANK YOU